Planning-Guided Failure-Triggered Recovery via Imitation Learning for Dynamic Social Navigation

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Abstract—Classical navigation stacks are reliable in routine mapped environments but can fail in reciprocal interactions with moving people. End-to-end learned navigation can address such interactions, yet replacing a mature planner everywhere changes its behavior even where no recovery is needed. We study a bounded alternative: the classical planner remains active normally, while an observable failure detector invokes a learned policy only during collision risk, freezing, oscillation, or deadlock. The policy selects among 21 temporary subgoals and four interpretable modes; a planning-derived mask rejects geometrically invalid actions, and Nav2 executes the selected goal. A privileged short-horizon expert labels simulator states, behavior cloning initializes the policy, and two rounds of DAgger cover learner-induced recovery states. In a preliminary repeated high-density crossing-flow pilot, classical DWB reached 0/5 goals with five collisions, whereas the safety-aligned recovery policy reached 3/5 goals with no collision and two timeouts. This same-seed pilot is not statistically conclusive ($p = 0.167$); it establishes closed-loop feasibility and motivates the locked multi-scenario evaluation rather than a robustness claim.

I. Introduction

Mobile robots operating around people inherit complementary weaknesses from classical and learned navigation. Classical stacks offer explicit maps, reusable planners, and predictable nominal behavior, but local optimization can freeze or oscillate in interactive bottlenecks. Learned policies can encode interaction patterns, but continuous end-to-end replacement expands the learned component's authority and makes failures harder to interpret.

We ask whether learning can be confined to the failure boundary. Figure 1 shows the resulting hierarchy. A classical planner owns nominal PointGoal navigation. An observable detector and hysteretic state machine trigger a recovery option. A compact policy then chooses an interpretable temporary target or mode from a fixed action set, subject to a mask shared by data generation and deployment. Once progress is restored, the original task goal is reissued.

The current contribution is threefold: (i) a failure-triggered architecture that preserves classical nominal control; (ii) an automated privileged planning expert that labels masked temporary-goal choices; and (iii) a deployment-aligned imitation pipeline using behavior cloning and two DAgger rounds. We do not claim a first recovery system or a formal safety guarantee. The present manuscript reports verified pilot evidence and clearly separates it from the pending final multi-seed evaluation.

II. Related Work

Arena-Bench and Arena-Rosnav provide reproducible interfaces for comparing model-based and learned navigation in dynamic environments [1], [2], [3], [4]. All-in-One learns to select among navigation planners [5]; DR-MPC instead learns a residual around model-predictive control [6]. Both motivate hybrid navigation but differ from a bounded failure-only intervention.

Learned navigation recovery predates this work. Del Duchetto et al. learn failure detection and local recovery from human demonstrations [7]. Mohammad et al. proactively predict planning failure with a Gaussian process and search for a recovery state [8]. Recent failure-aware learning uses unsafe navigation experience to shape value estimation while restricting policy learning to successful demonstrations [9]. Our narrower distinction is to generate recovery demonstrations automatically by privileged rollout planning, express decisions as temporary subgoals, and aggregate labels on the recovery states induced by the learner. DAgger provides the standard mechanism for mitigating this sequential distribution shift [10].

III. Method

A. Failure-Triggered Hierarchy

Let g be the original global goal and let the classical stack produce command u_b. From observable history o_t, the detector returns four scores for collision risk, freeze, oscillation, and deadlock. Recovery begins after the maximum score exceeds τ_{on} and exits only after it remains below $\tau_{\text{off}} < \tau_{\text{on}}$ while valid goal progress resumes. Cooldown, option duration, and recovery-count limits prevent rapid switching.

B. Action Space and Mask

The action space contains temporary goals at radii $\{0.6, 1.0, 1.4\}$ m and bearings $\{-90, -60, -30, 0, 30, 60, 90\}^\circ$, plus WAIT, BACKUP, REPLAN, and CONTINUE. The mask rejects occupied, disconnected, LiDAR-occluded, rear-unobserved, or path-divergent motion. During latched collision risk it also blocks CONTINUE and REPLAN and applies the combined robot–human clearance to candidate endpoints and swept segments. Masking is applied before softmax, so invalid action rate is zero by construction when at least one fallback remains.

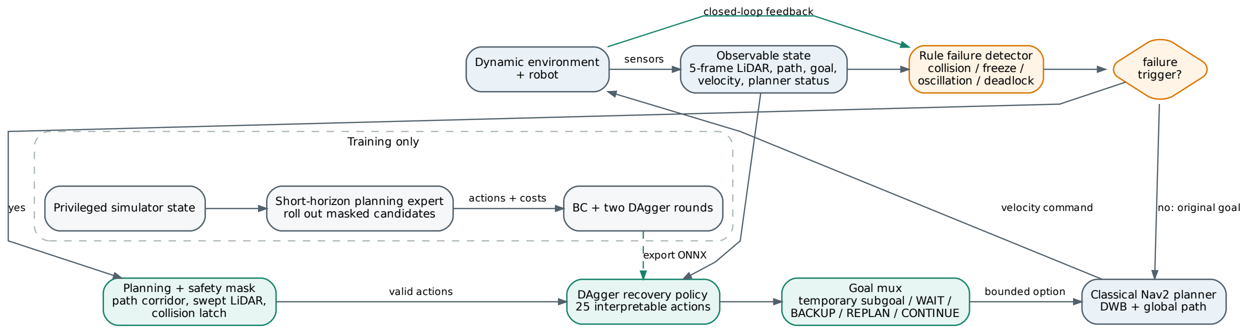

Fig. 1. Closed-loop architecture. Privileged state and expert rollouts are used only during training. At test time, the recovery policy receives observable state and can only choose actions admitted by the planning and safety mask.

An independent stopping layer uses

$$d_{\text{stop}} = \frac{v^2}{2a_{\text{brake}}} + v\ell + m, \tag{1}$$

where ℓ is control latency and m is an empirical margin. It may override learning but is not a formal collision-avoidance certificate.

C. Privileged Expert and Imitation

During training only, the expert receives the static grid and true pedestrian states. Each legal candidate is rolled out for a short horizon using differential-drive kinematics and predicted human motion. It minimizes collision, progress, social distance, path rejoin, length, smoothness, time, and action-switch costs; collision and masked actions receive hard penalties. The selected action and all candidate costs are stored with the deployable observation.

The policy uses a one-dimensional CNN over five 180-beam scans and an MLP over goal, local path, velocity, base command, temporal progress, planner state, and detector scores. Behavior cloning minimizes cross entropy to the expert. DAgger then executes the current policy only on training scenarios, queries the privileged expert on visited recovery states, aggregates the results, and retrains. Two rounds are completed; model selection uses scenario-disjoint validation rather than iteration count.

IV. Experimental Setup

Experiments use ROS2 Humble, Nav2 DWB, and the pinned Arena Humble Gazebo profile. The installed profile could not provide a complete HuNav dependency chain, so deterministic LiDAR-visible kinematic pedestrian proxies are used and disclosed as a fallback. The robot receives a known static map and a 270-degree planar LiDAR scan downsampled to 180 beams. Train, validation, and test splits are scenario-separated.

TABLE I
Repeated high-density crossing-flow pilot. The same scenario seed is repeated to expose simulator scheduling variance; these data are not the final multi-seed evaluation.

Method	Goal	Collision	Timeout	$d_{\min}$ [m]	Time [s]
Classical DWB	0/5	5/5	0/5	0.708	–
Triggered DAgger	3/5	0/5	2/5	1.080	110.4

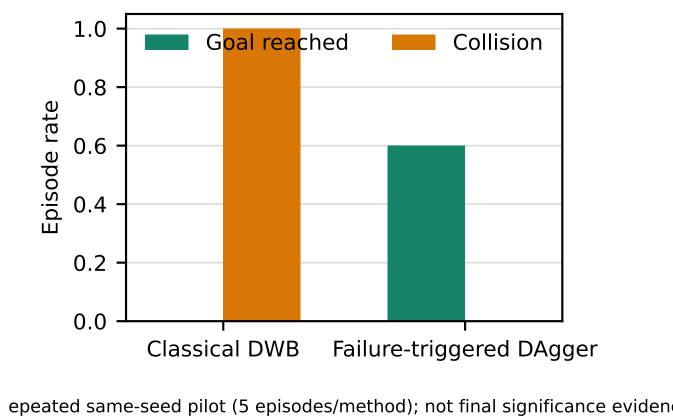

Fig. 2. Pilot outcome rates. Exact intervals are wide and the two-sided unpaired Fisher test for success gives $p = 0.167$.

The current pilot repeats one locked high-density crossing-flow validation seed five times per method to expose residual Gazebo scheduling variance. The episode horizon is 180 s. Outcomes include every valid reset; simulator/reset failures are reported separately and never converted into algorithm failures. This pilot is a development gate, not the final evaluation.

V. Pilot Results and Analysis

Table I and Fig. 2 show a concrete closed-loop signal: the selected policy executes WAIT, BACKUP, and temporary subgoals, rejoins the original goal, and removes collisions in these five repeats. Three episodes reach the goal.

Two time out, including one episode stopped beside a lateral LiDAR return inconsistent with both scenario geometry and privileged pedestrian positions. We retain both timeouts. The sample is too small for a statistical improvement claim, and the different navigation-time tail remains a central tradeoff for expanded evaluation.

A separate retained medium-density counterexample exposed a safety-controller limit cycle: the ordinary swept capsule rejected translation because its start footprint was already within a rear obstacle's conservative clearance. Requiring emergency translation to increase distance from every initial overlap repaired the exact replay (goal reached in 114.4 s instead of a 180 s timeout), but classical DWB remained faster at 100.9 s. We therefore treat this as anti-regression evidence rather than learned-policy superiority.

Offline scenario-disjoint validation of the selected checkpoint gives 90.98% top-1, 98.50% top-3, 95.49% near-optimal actions, zero invalid selections, and zero catastrophic selections under the expert cost. These metrics validate deployment alignment but cannot replace closed-loop navigation outcomes.

VI. Limitations

The method currently assumes a known two-dimensional map and planar LiDAR. Its privileged expert depends on simulator state and constant-velocity pedestrian prediction. The finite discrete action set cannot express arbitrary maneuvers. The Gazebo kinematic fallback differs from real crowds and exhibits scheduling and proxy artifacts. There is no formal safety guarantee, no hardware study, and no completed learned detector or PPO refinement. The reported pilot repeats one scenario seed and is not the final multi-scenario test.

VII. Conclusion

Failure-triggered temporary-goal recovery can be integrated with a classical ROS2 navigation stack and trained from an automated privileged planner without granting the learned policy continuous velocity authority. The selected DAgger checkpoint demonstrates real recovery and goal rejoin in a difficult crossing-flow pilot while eliminating observed collisions, but its confidence interval and timeout tail preclude broad claims. The next required evidence is a locked, paired, multi-scenario and multi-seed evaluation.

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